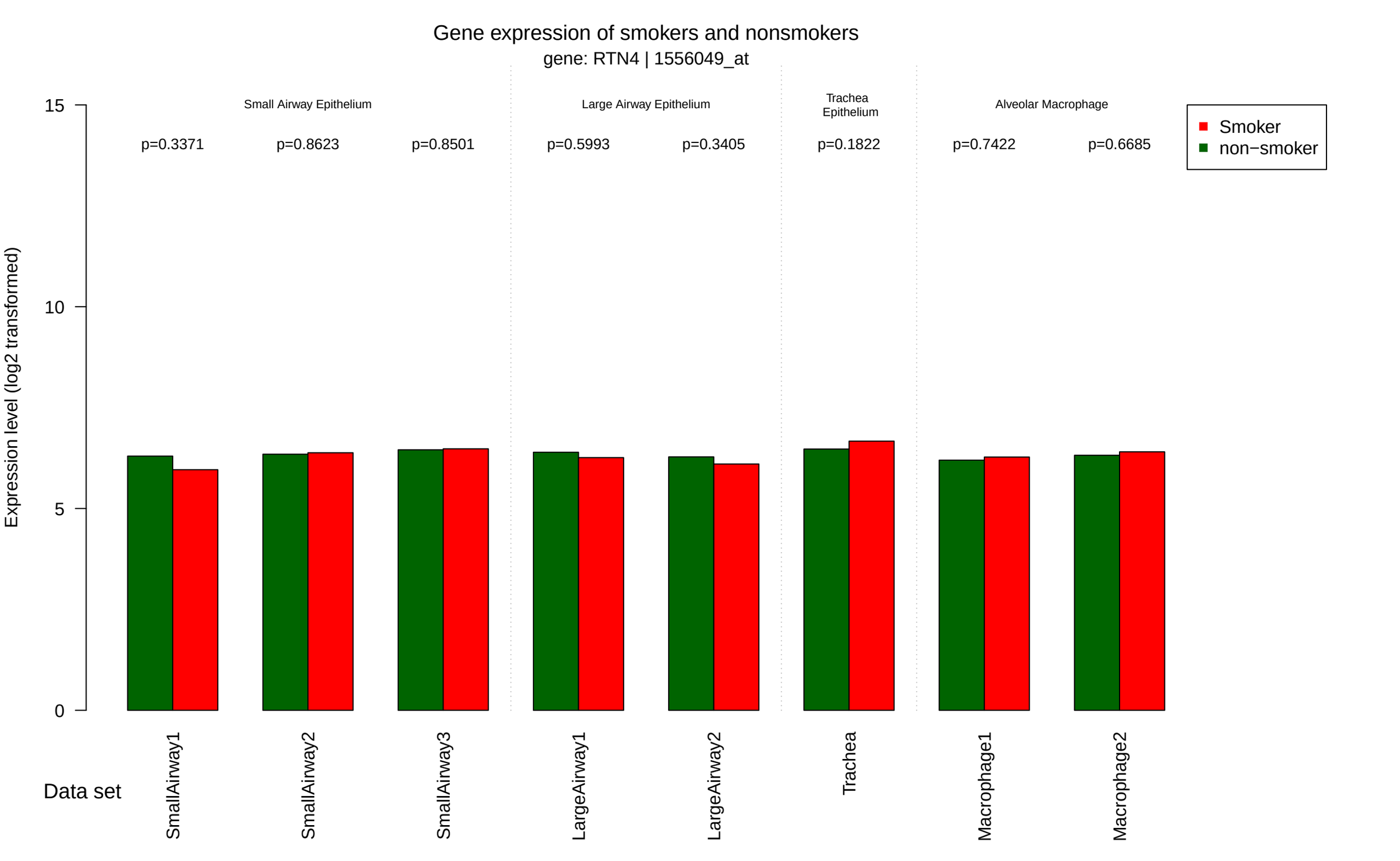
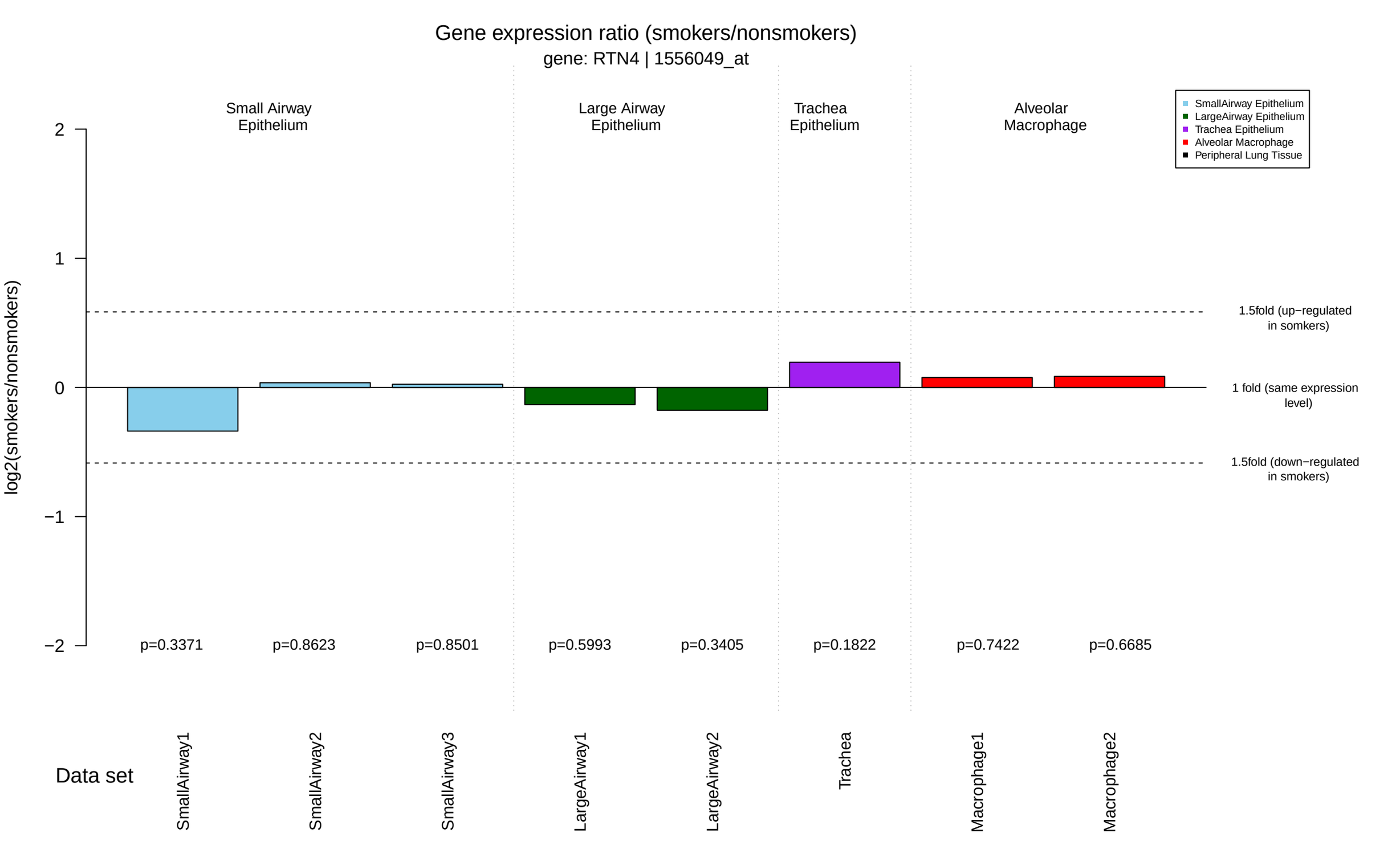
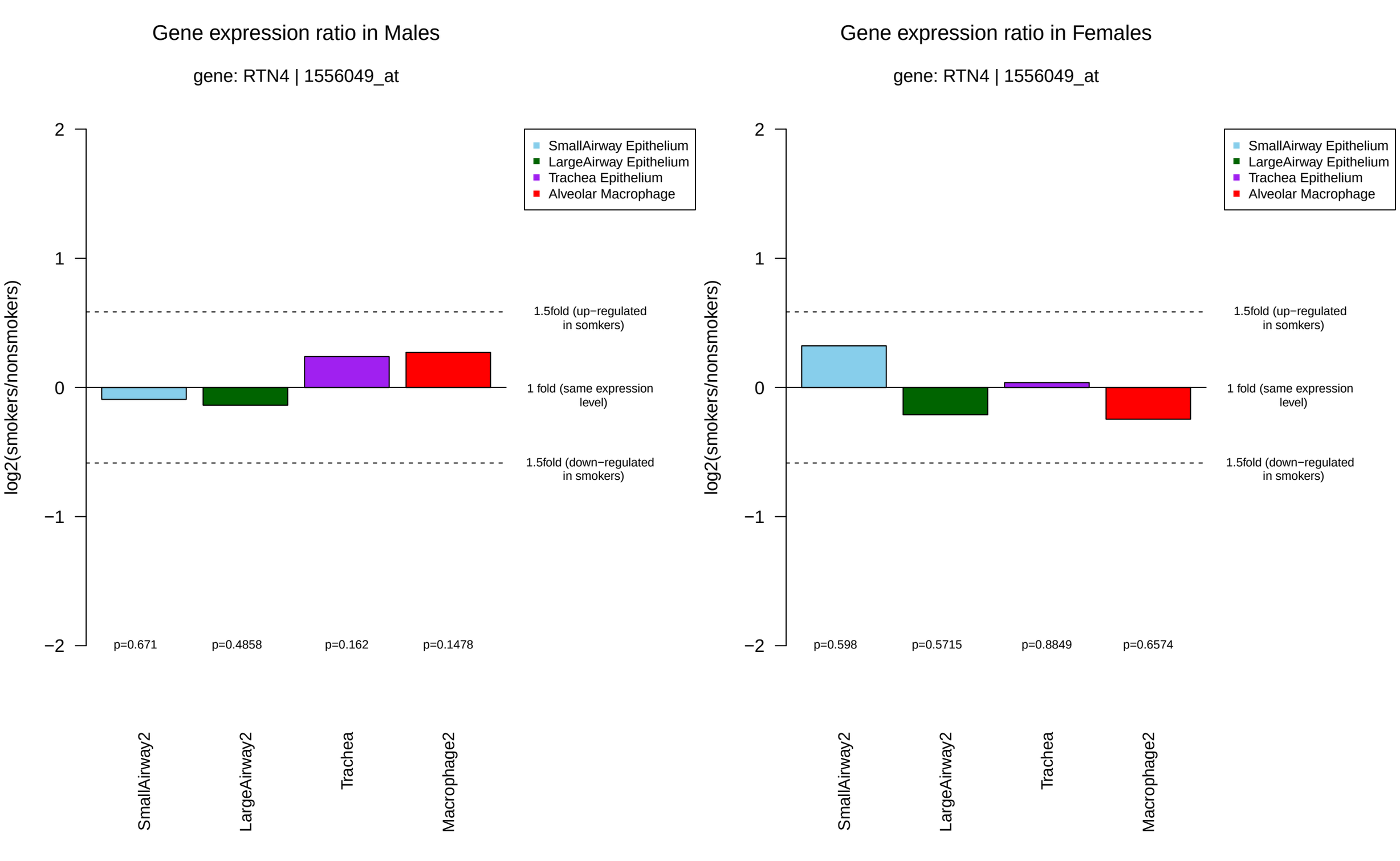
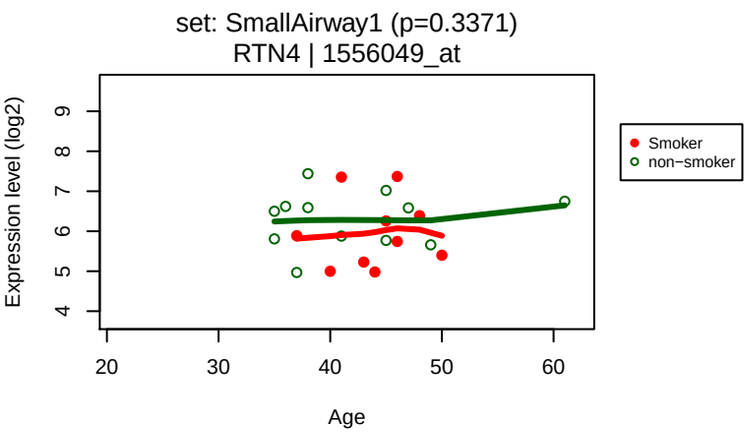
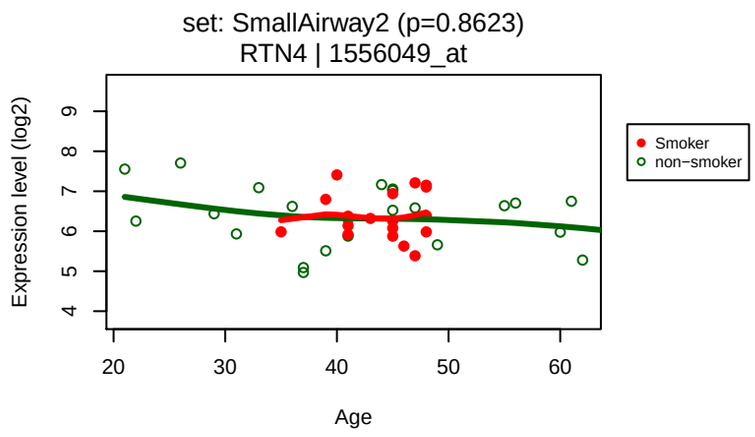
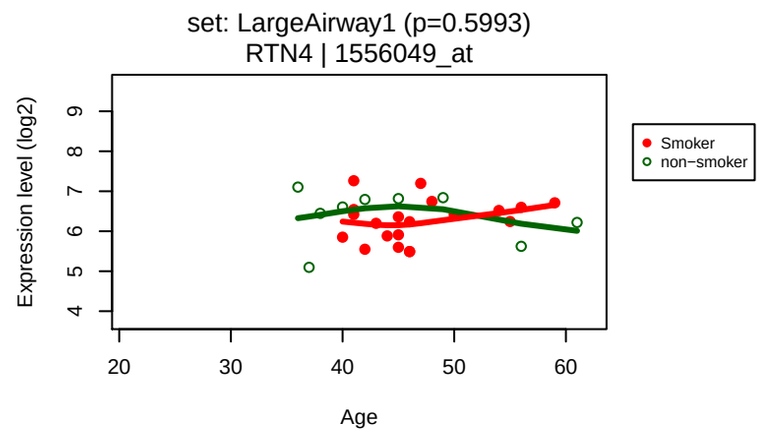

Gene expression of smokers and nonsmokers
gene: RTN4 | 1556049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3371
p=0.8623
p=0.8501
p=0.5993
p=0.3405
p=0.1822
p=0.7422
p=0.6685
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

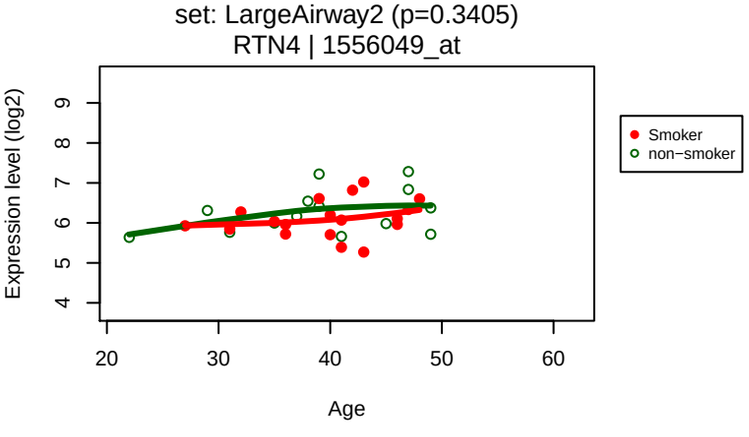

Gene expression ratio (smokers/nonsmokers)
gene: RTN4 | 1556049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3371
p=0.8623
p=0.8501
p=0.5993
p=0.3405
p=0.1822
p=0.7422
p=0.6685
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

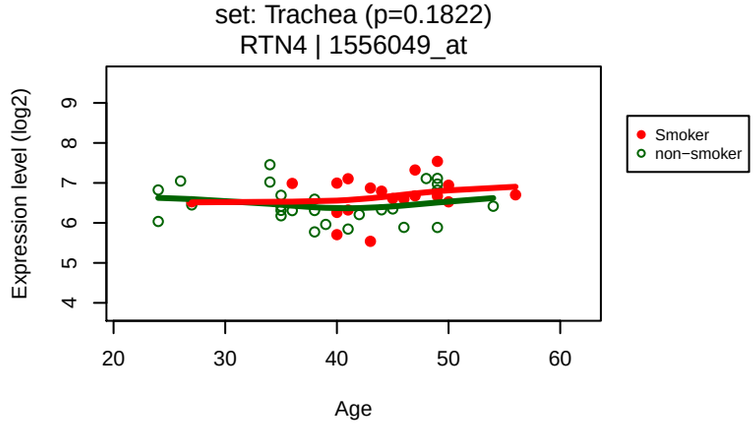

Gene expression ratio in Males
gene: RTN4 | 1556049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.671
p=0.4858
p=0.162
p=0.1478
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTN4 | 1556049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.598
p=0.5715
p=0.8849
p=0.6574
SmallAirway2
LargeAirway2
Trachea
Macrophage2

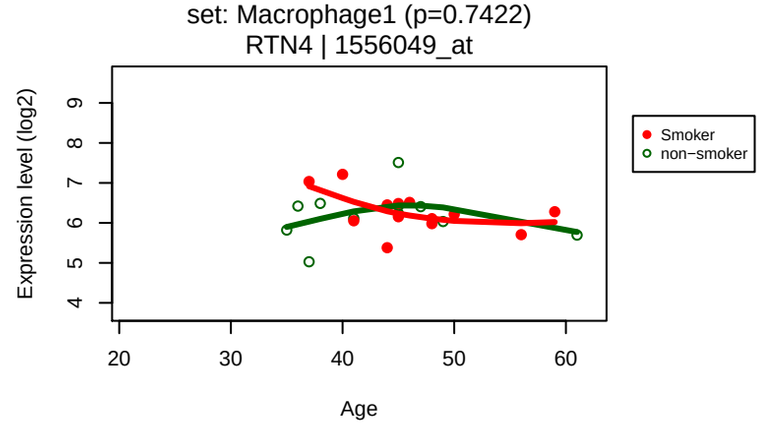

set: SmallAirway1 (p=0.3371)
RTN4 | 1556049_at
Expression level (log2)
Age
Smoker
non−smoker

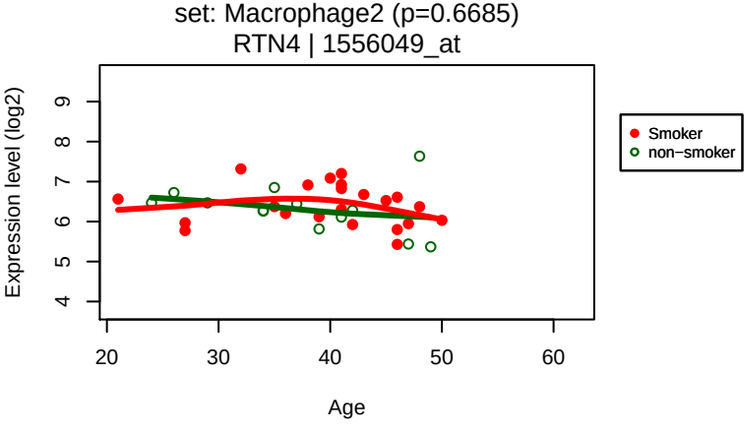

set: SmallAirway2 (p=0.8623)
RTN4 | 1556049_at
Expression level (log2)
Age
Smoker
non−smoker

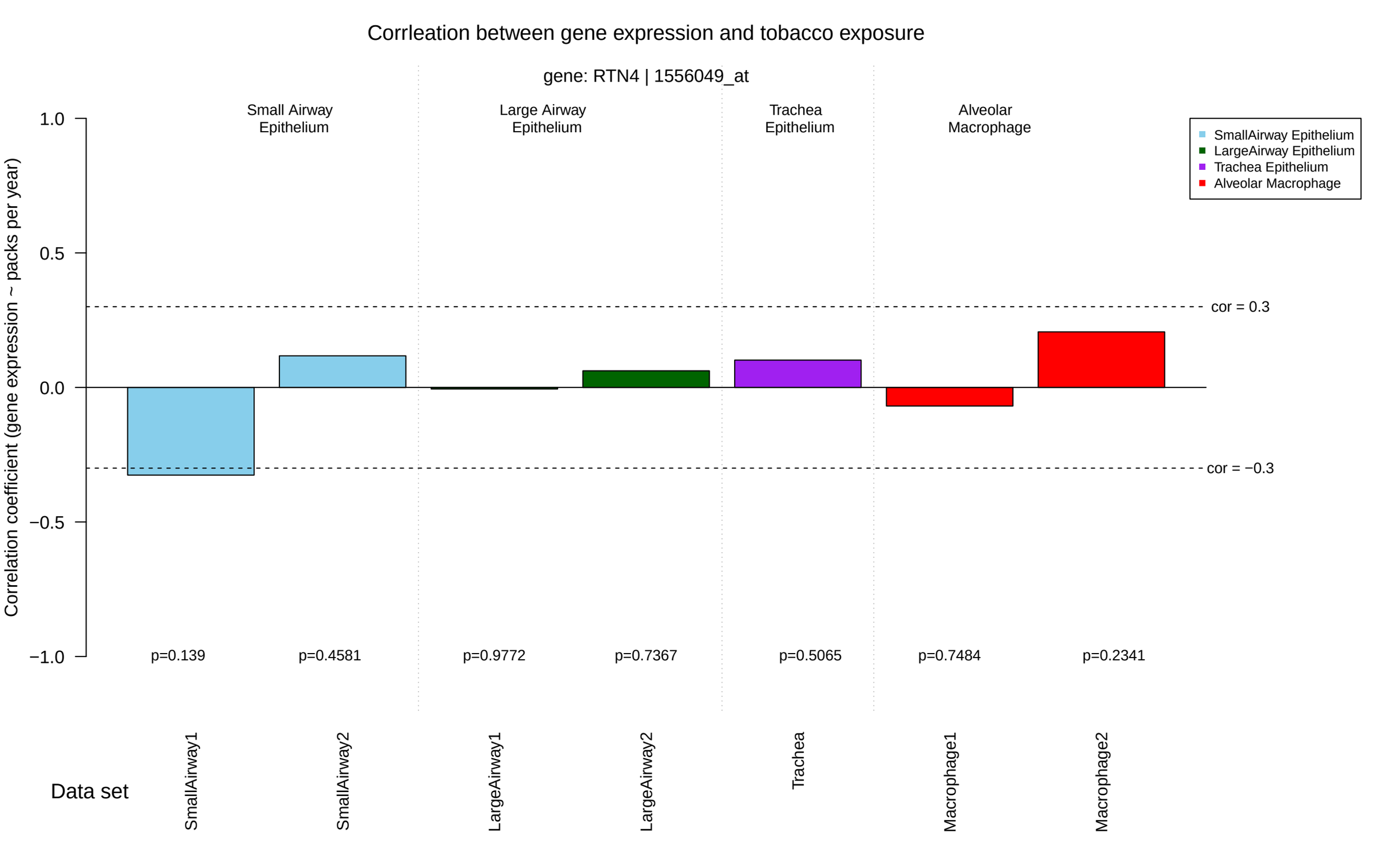

set: LargeAirway1 (p=0.5993)
RTN4 | 1556049_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3405)
RTN4 | 1556049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1822)
RTN4 | 1556049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7422)
RTN4 | 1556049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6685)
RTN4 | 1556049_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTN4 | 1556049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.139
p=0.4581
p=0.9772
p=0.7367
p=0.5065
p=0.7484
p=0.2341
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2